Case study: Clavibacter

Prepare session:

```
library(ggtree)
library(treeio)
library(tidyverse)
```

```
# Load a tree file
tree <- read.newick("example_phylo.nwk")
tree$node.label <- round(as.numeric(tree$node.label), digits = 3)
```

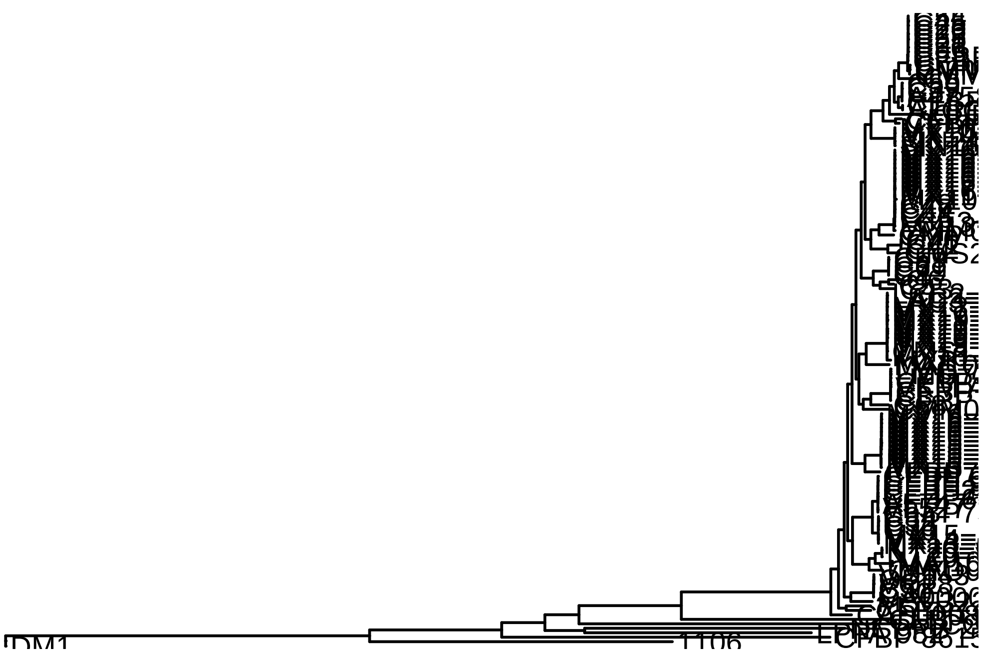
Prepare metadata

```
# Cargar metadata
metadata <- readr::read_tsv("Experiment_Design_example_1.tsv")

metadata <- metadata %>%
  rename(label = Assembly.Accession)  # Para coincidir con etiquetas del árbol

# Unir metadata con etiquetas del árbol
metadata_ordered <- metadata %>%
  filter(label %in% tree$tip.label) %>%
  arrange(match(label, tree$tip.label))


metadata_ordered <- metadata_ordered[,c(-1)]
tree$tip.label <- metadata_ordered$Strain

matrix_heatmap <- data.frame(metadata_ordered %>%
```

```
  mutate(Country = str_extract(Location, "^[^:]+")) %>%
  select(Country))

metadata_ordered <- metadata_ordered %>%
  mutate(Country = str_extract(Location, "^[^:]+"))

rownames(matrix_heatmap) <- make.unique(metadata_ordered$Strain)
rownames(metadata_ordered) <- make.unique(metadata_ordered$Strain)

country_colors <- RColorBrewer::brewer.pal(n = min(length(unique(matrix_heatmap$Country)), 3
```

Graph

```
library(phytools)
# reroot

# Reenraizar en la hoja "LeafName"
tree_rerooted <- reroot(tree, node = which(tree$tip.label == "DM1"))

ggtree(tree_rerooted) %<+% metadata_ordered +
  geom_tiplab(aes(label = label))
```



```r
g<-ggtree(tree_rerooted, ladderize = TRUE, layout = "circular") %<+% metadata_ordered +
  geom_tiplab(aes(label = label, angle = angle),
              linesize = .1, size = 5, offset = .01) +
  geom_nodelab(size = 4, color = "black") +
  geom_tippoint(aes(color = Country), size = 8, shape = 20) +
  geom_tippoint(aes(shape = Source), size = 10, alpha = 0.5) +
  scale_shape_manual(values = c(18, 8, 18)) +
  labs(shape = "Source")

# Generar el grafico
p4 <- ggtree(tree_rerooted, ladderize = TRUE, layout = "circular") %<+% metadata_ordered +
  geom_tiplab(aes(label = label, angle = angle), linesize=.1, size = 8, align = FALSE, color
              offset = 1.5) +
  geom_nodelab(size = 4, color = "black", nudge_x = 0.1) +
  geom_tippoint(aes(color = Country), size = 8, shape = 20) +
  geom_tippoint(aes(shape = Source), size = 10, alpha = 0.5) +
  scale_shape_manual(values = c(18, 8, 18)) +
  labs(shape = "Source")
# p4
```